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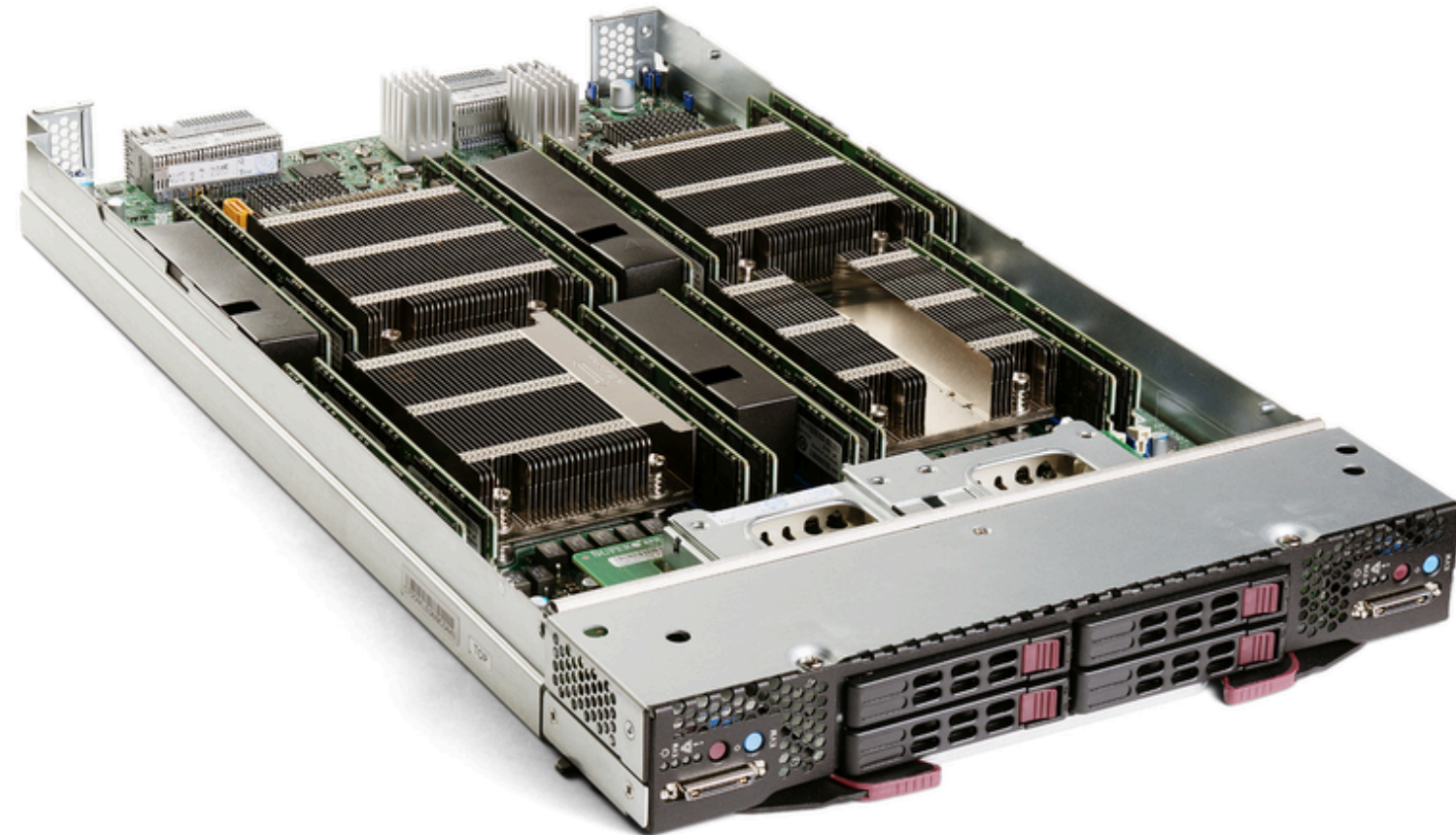
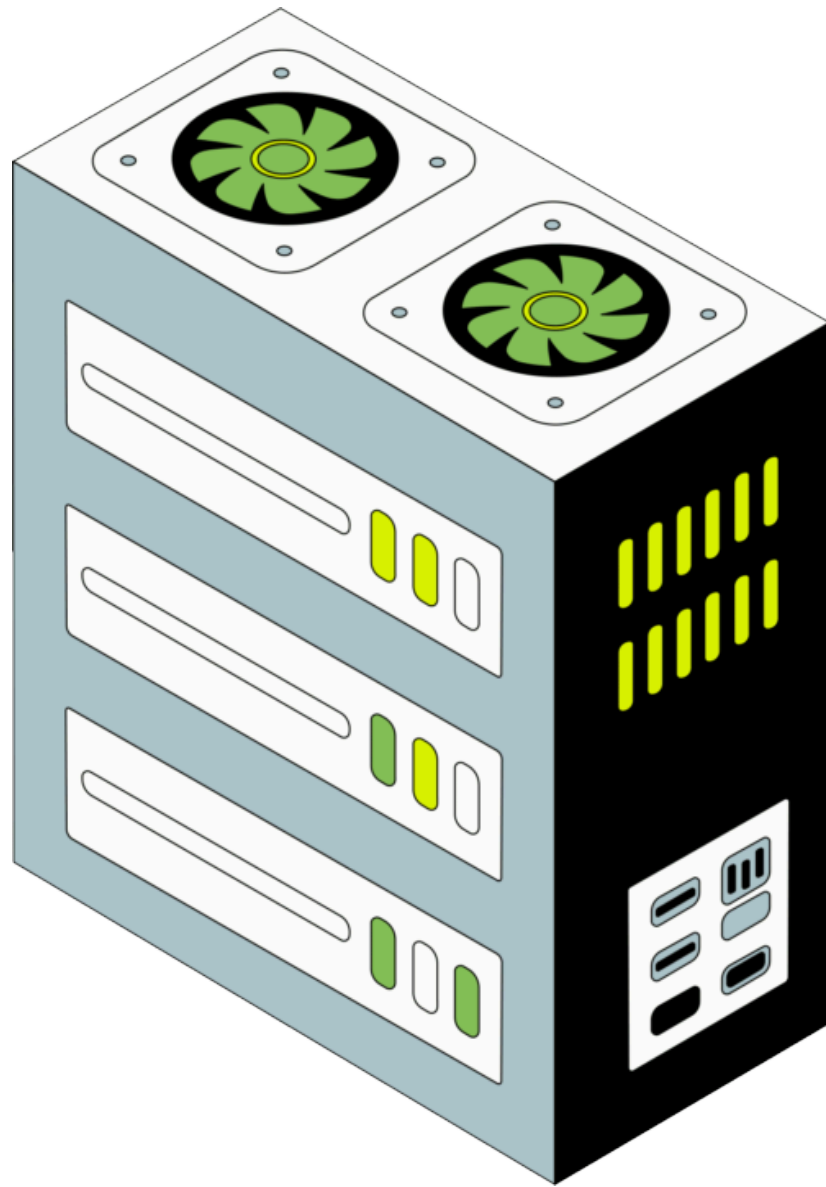
Introduction

CSAI321: Computer Networks

Join the lecture online on your dashboard

What is the Cloud?

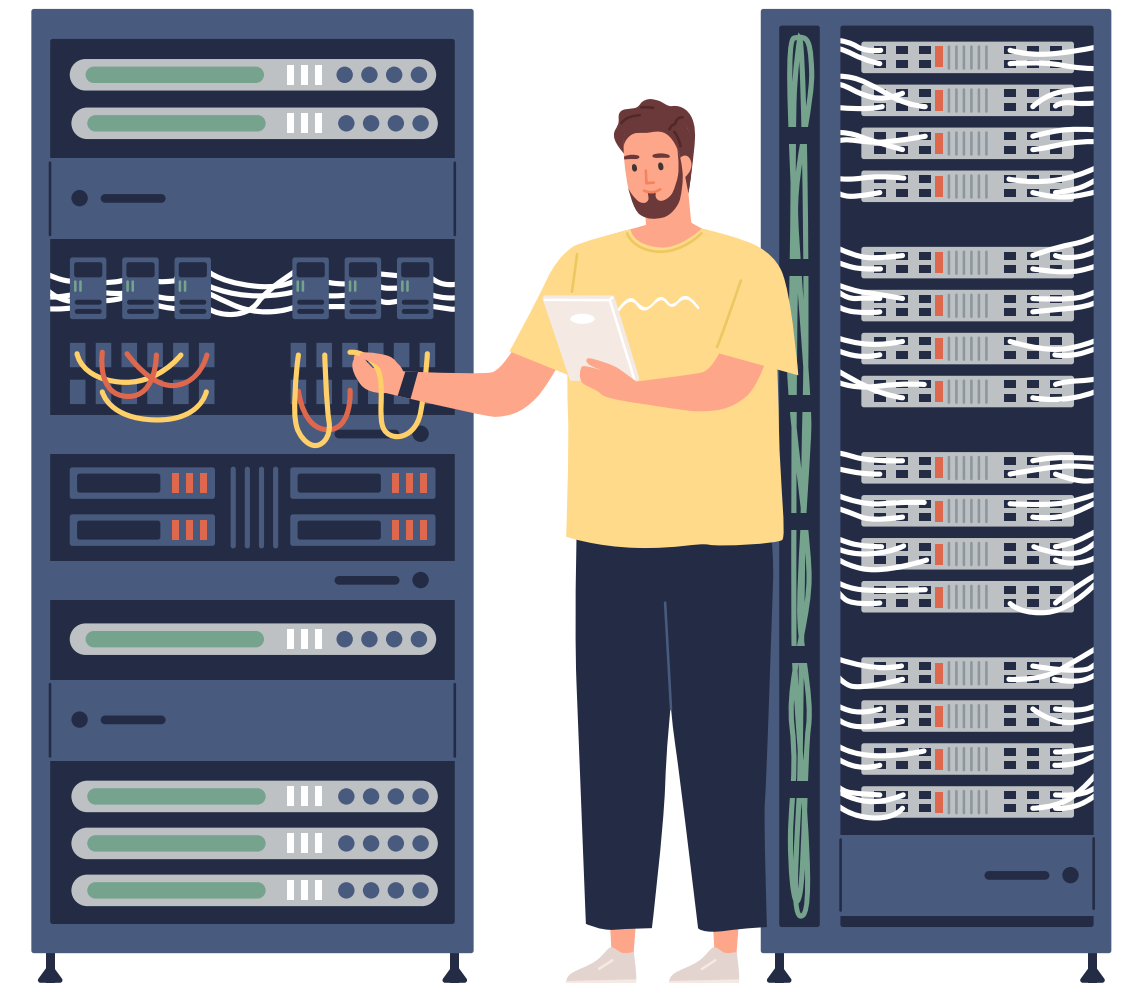
Is the cloud just someone else's computer?



What is the Cloud?



More like a lot of computers...

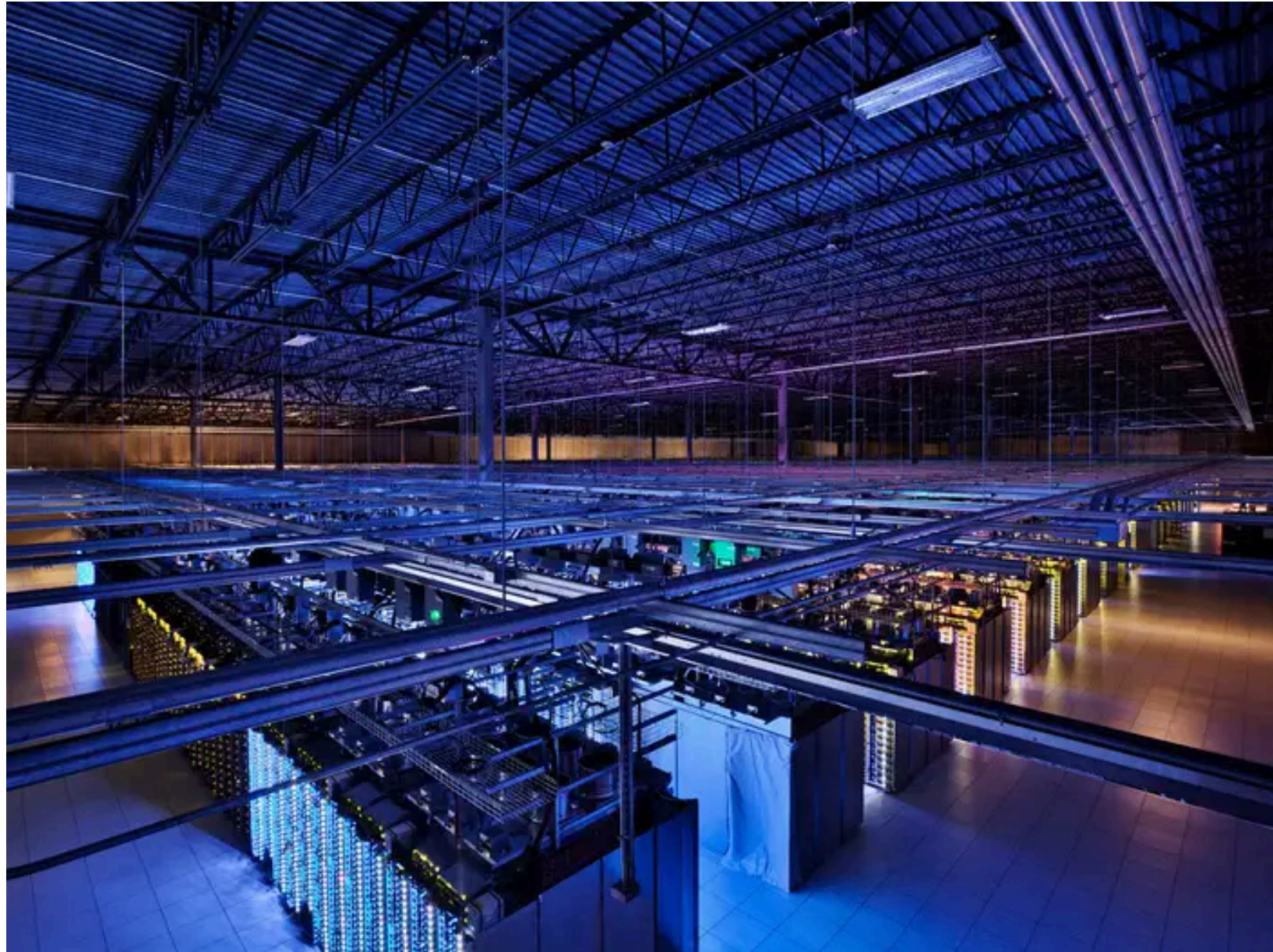


What is the Cloud?



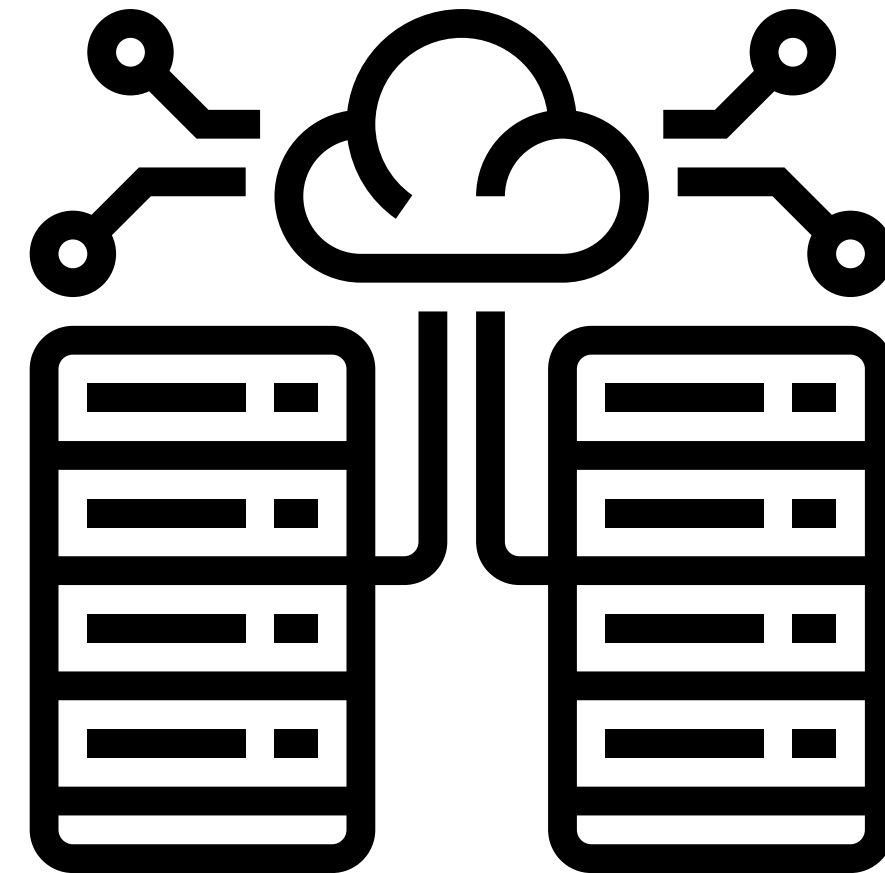
A lot, lot!

What is the Cloud?



An overhead view of the server infrastructure in Google's data center in Council Bluffs, Iowa.

And all these servers need to
talk to each other!



The cloud computing service providers – Amazon Web Services(AWS), Azure, Google Cloud Platform(GCP) all have super fast computer network



Getting all those 0s and 1s to you efficiently!

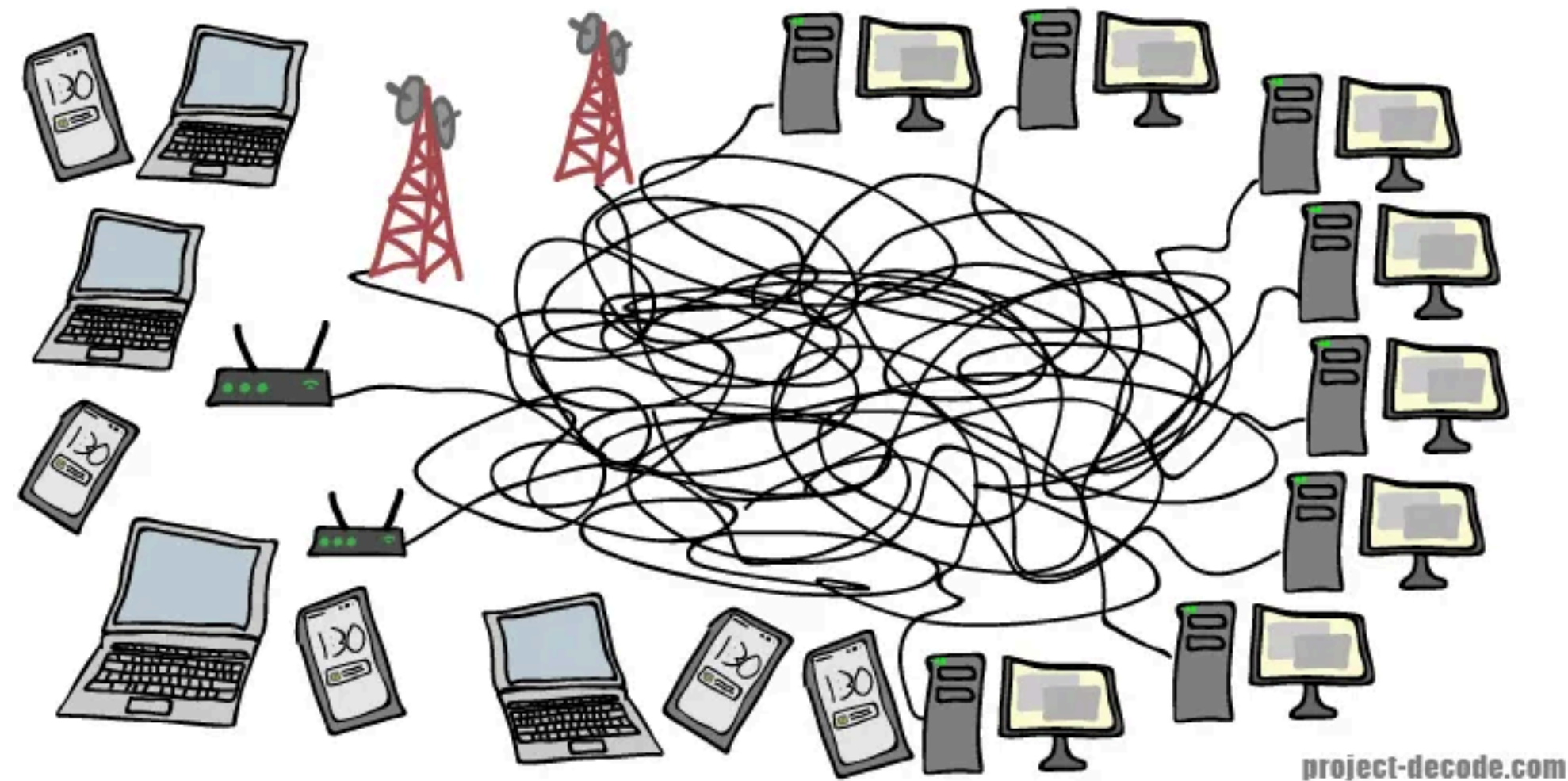
But then networks are not new

In 1960s, the UCLA computer tried to send “login,” to computer in Stanford Research Institute. Only “lo” got through before the system crashed.



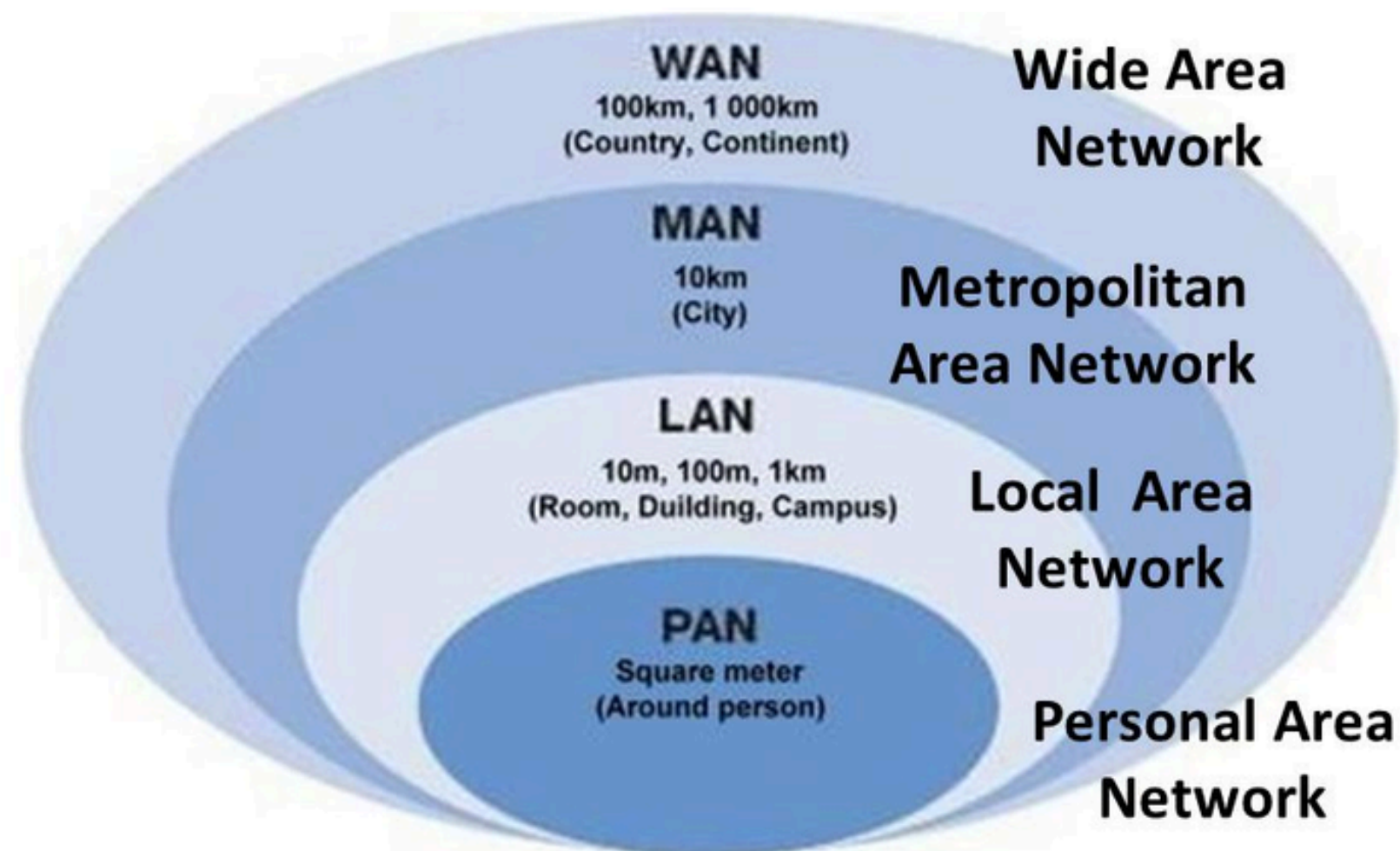
Okay, so computer networks are not new...

A Somewhat Accurate Drawing of the Internet



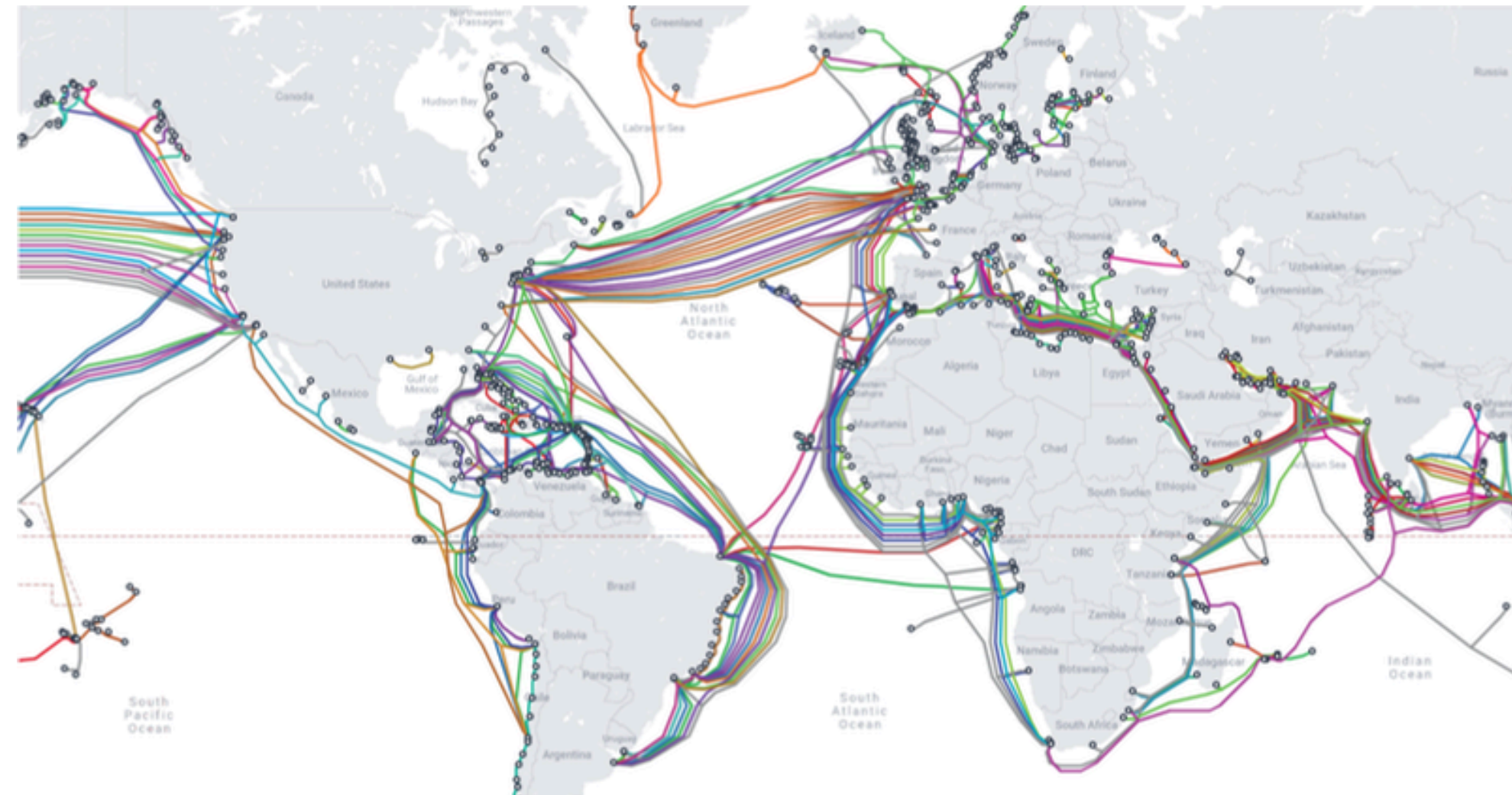
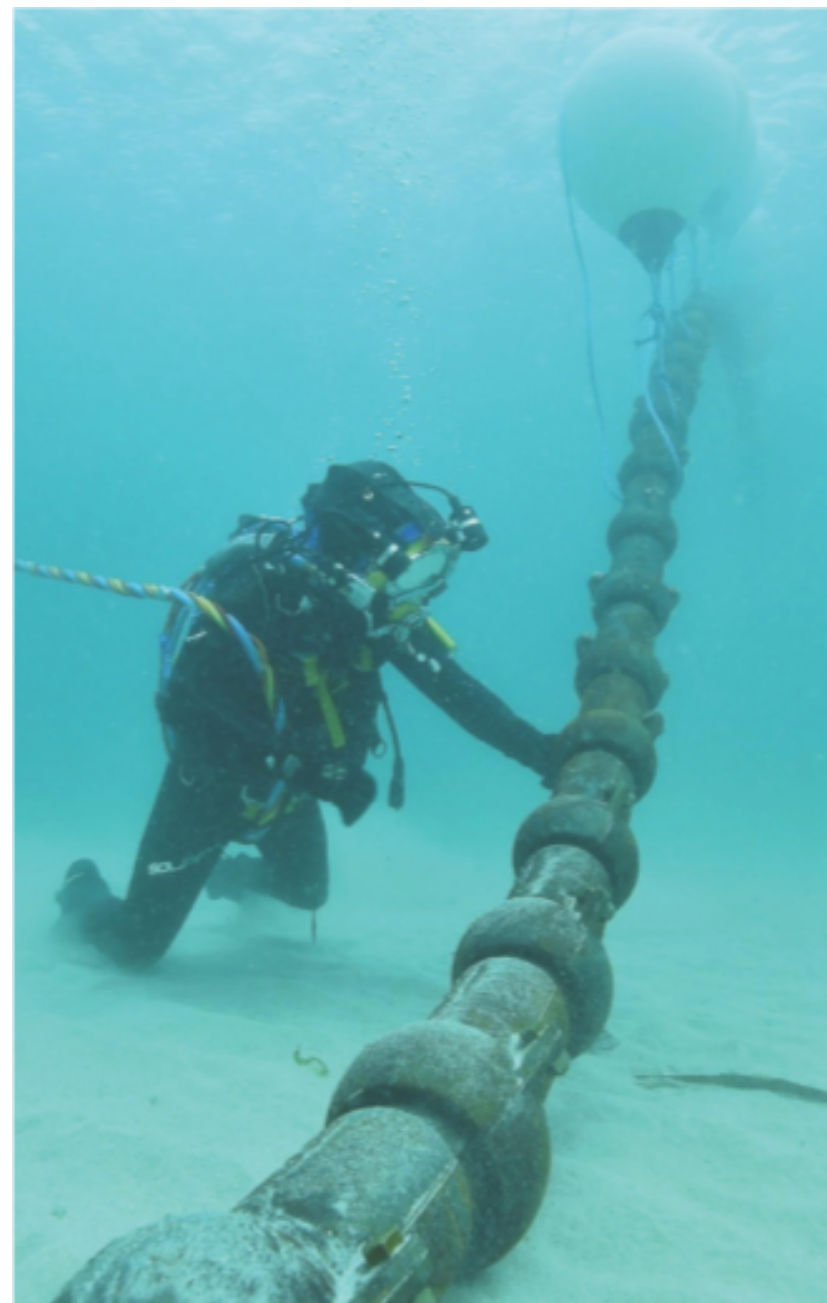
Traditionally, we have had...

Geographical Scope



- **WAN (Wide Area Network):** Covers larger geographic areas, like connecting different branch offices of a company, or even the internet itself.
- **MAN (Metropolitan Area Network):** Usually spans a city, often used by municipalities or large organizations.
- **LAN (Local Area Network):** Typically used in homes or offices to connect computers and devices within a limited area.
- **PAN (Personal Area Network):** Very short range, like connecting your smartphone to your smartwatch or wireless headphones.

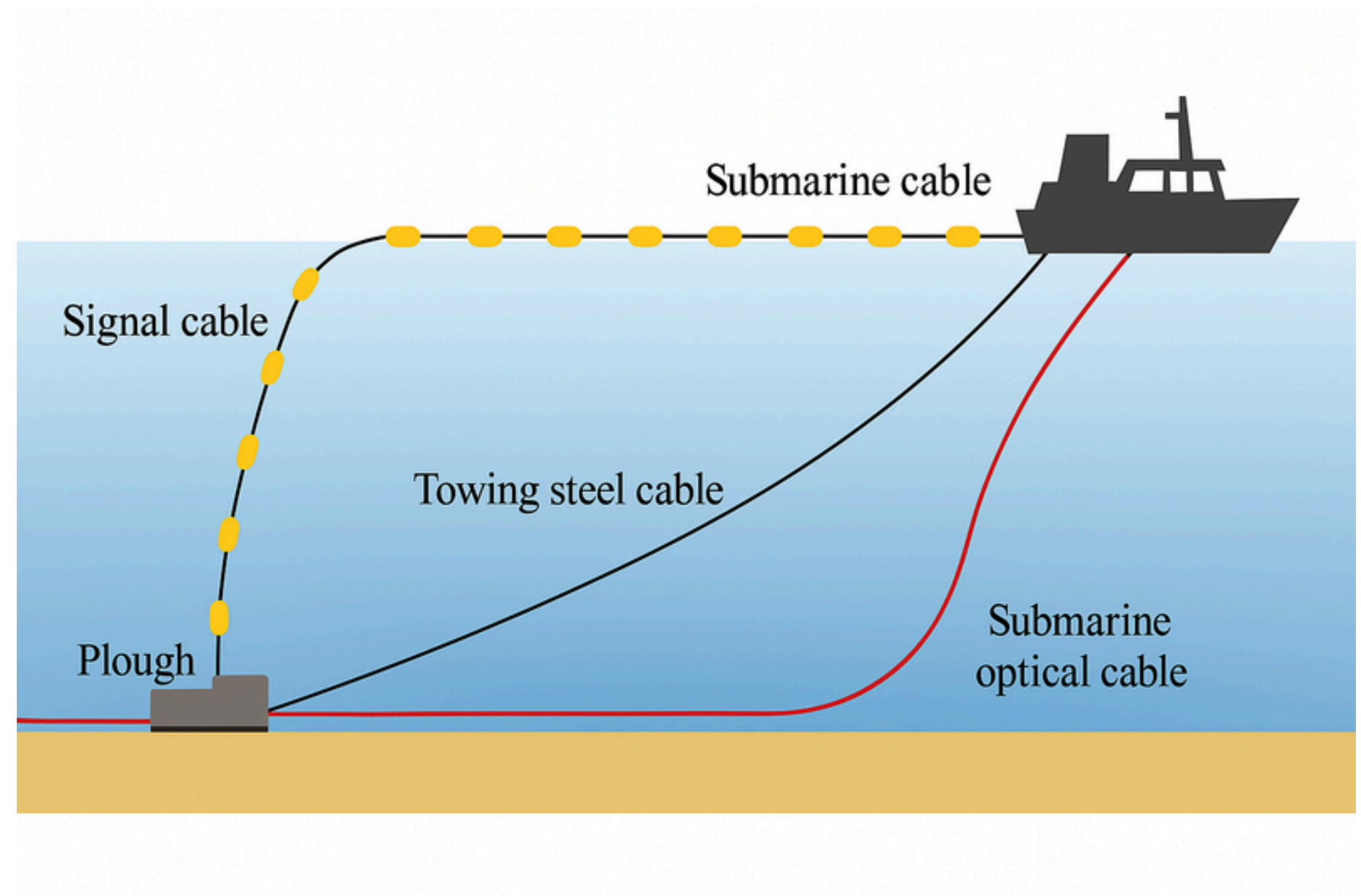
How extensive is the internet in the world?



Despite being called "the cloud," the internet relies on massive physical infrastructure spanning the globe. 597 cable systems in 2025

Why is it hard to keep them safe?

Any guesses?



Fishing anchors...

The major cause of transatlantic network traffic issues are fishing anchors mistakenly damaging the cables



Sharks trying to attack the cables!



What happens when?

What happens when you tap Instagram icon?

You tap the Instagram icon on your phone and a reel plays in under a second. List every 'thing' your data touched between your thumb and the video.

The Gap:

Most people stop at "WiFi → Instagram." The massive, intricate gap between that simple answer and reality is exactly what we study in this course.



NETFLIX

It's 10 PM. Millions are settling in for a Netflix movie. Suddenly, the app starts buffering or fails. What's happening behind the scenes?

Objective of the course

*We are going to demystify how networks work
Because networks are everywhere!*



Top-Down Approach.

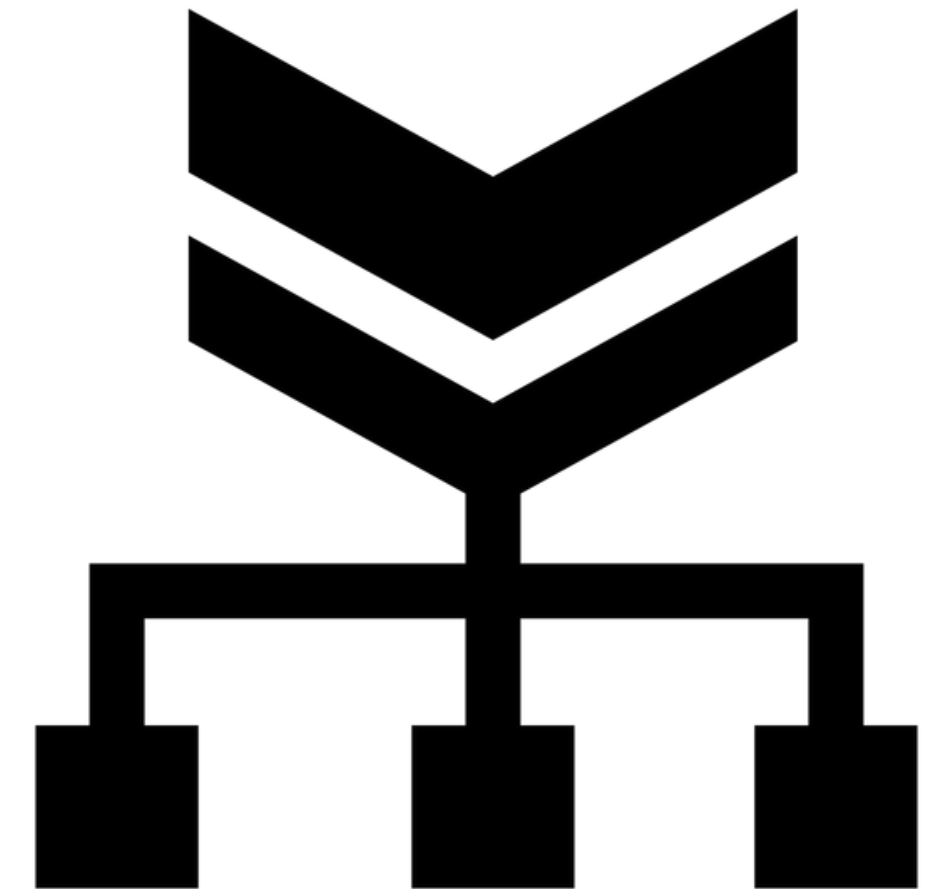
Why the Top-Down Approach?

Applications First

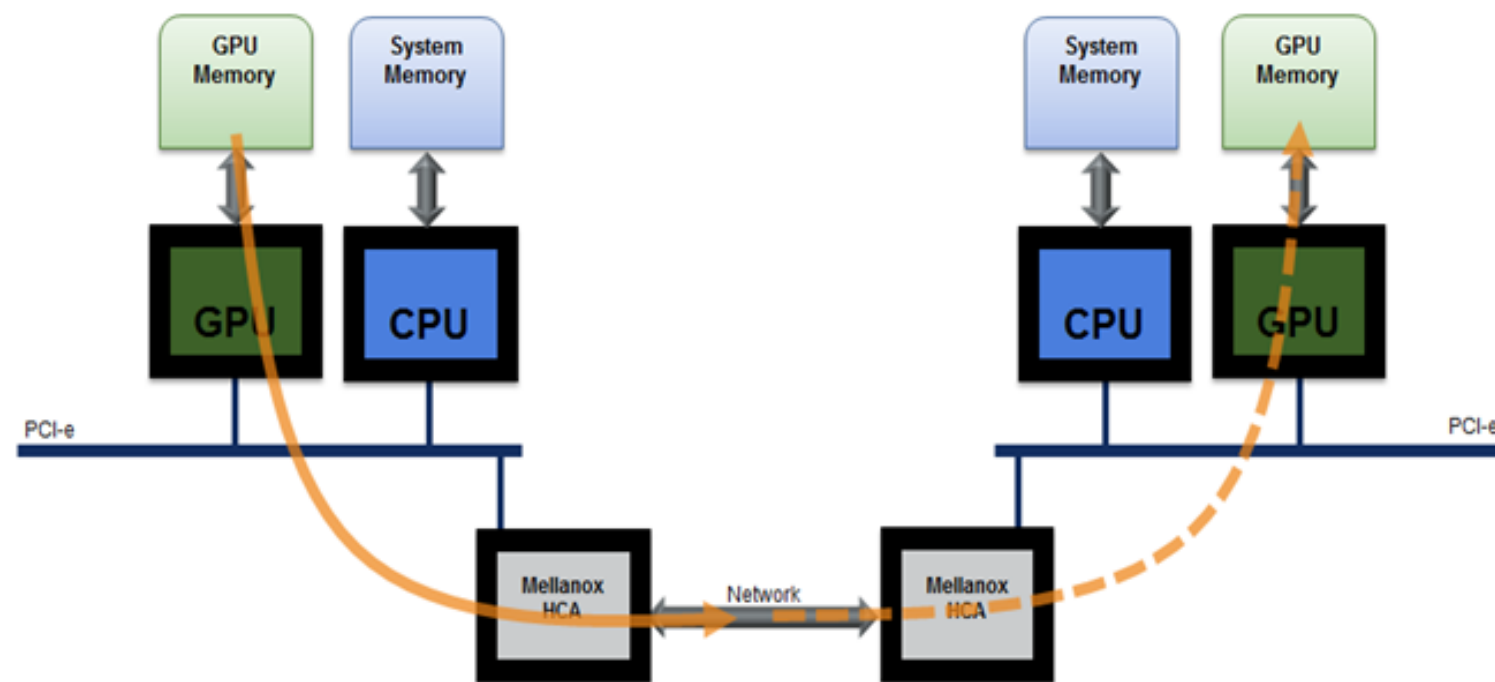
We start with what you know and use daily (Web, HTTP, DNS, Streaming). Understanding the requirements of these applications explains why we need the underlying network services.

Drilling Down to Physical

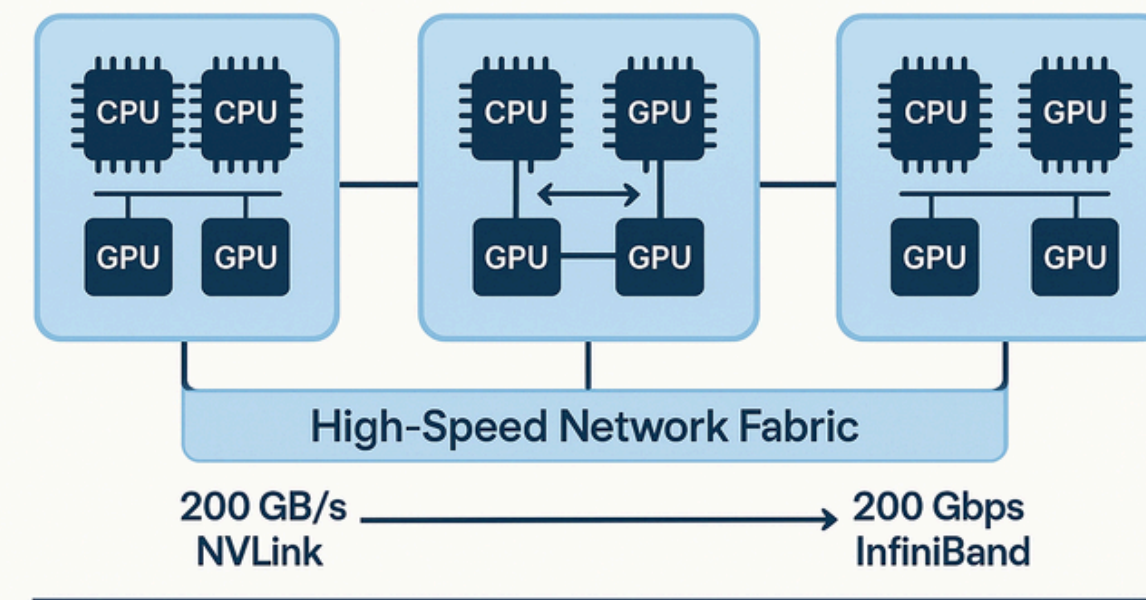
Once the application logic is clear, we progress downward through the Transport (TCP/UDP), Network (IP/Routing), Link, and Physical layers. This approach constantly answers "why" before "what".



Of course, GPUs are connected using very fast network cards!

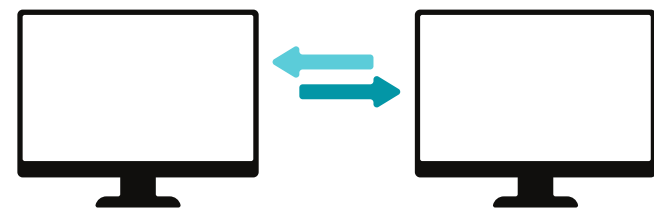


Networking is the backbone of GPU clusters

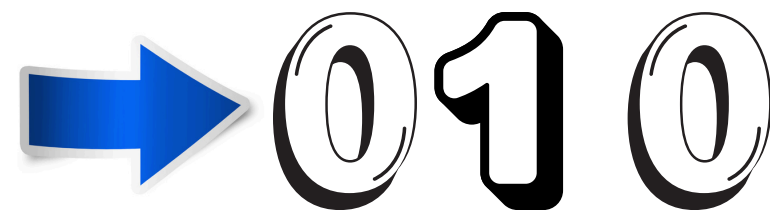


What to Expect from the Networking Course

Foundation

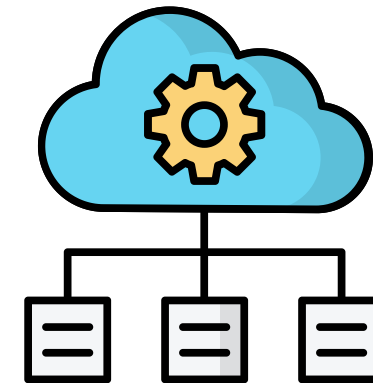


How computing
devices talk to each
other



How data travels

Hands-on Labs



Configure your own
network



Network Security

Join us on the adventure to discover how computer networks work

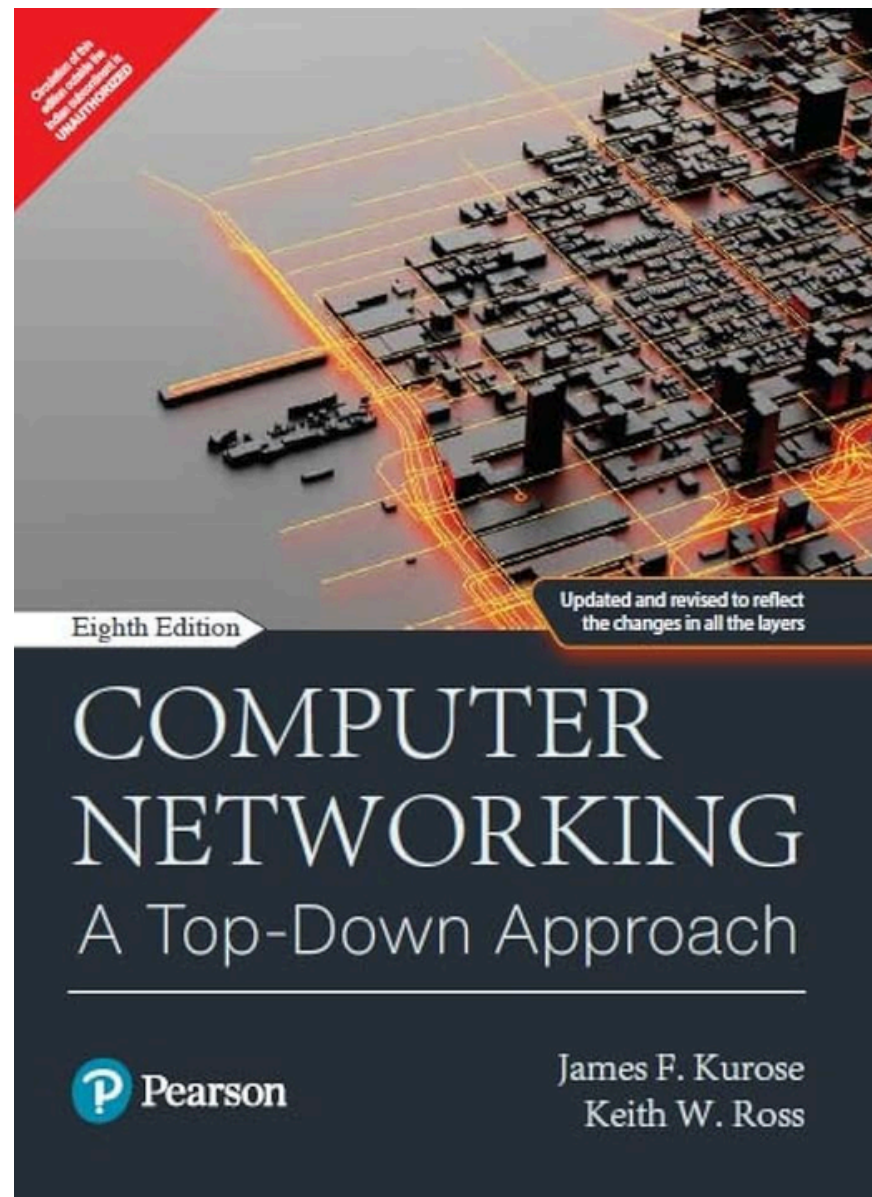


Grading

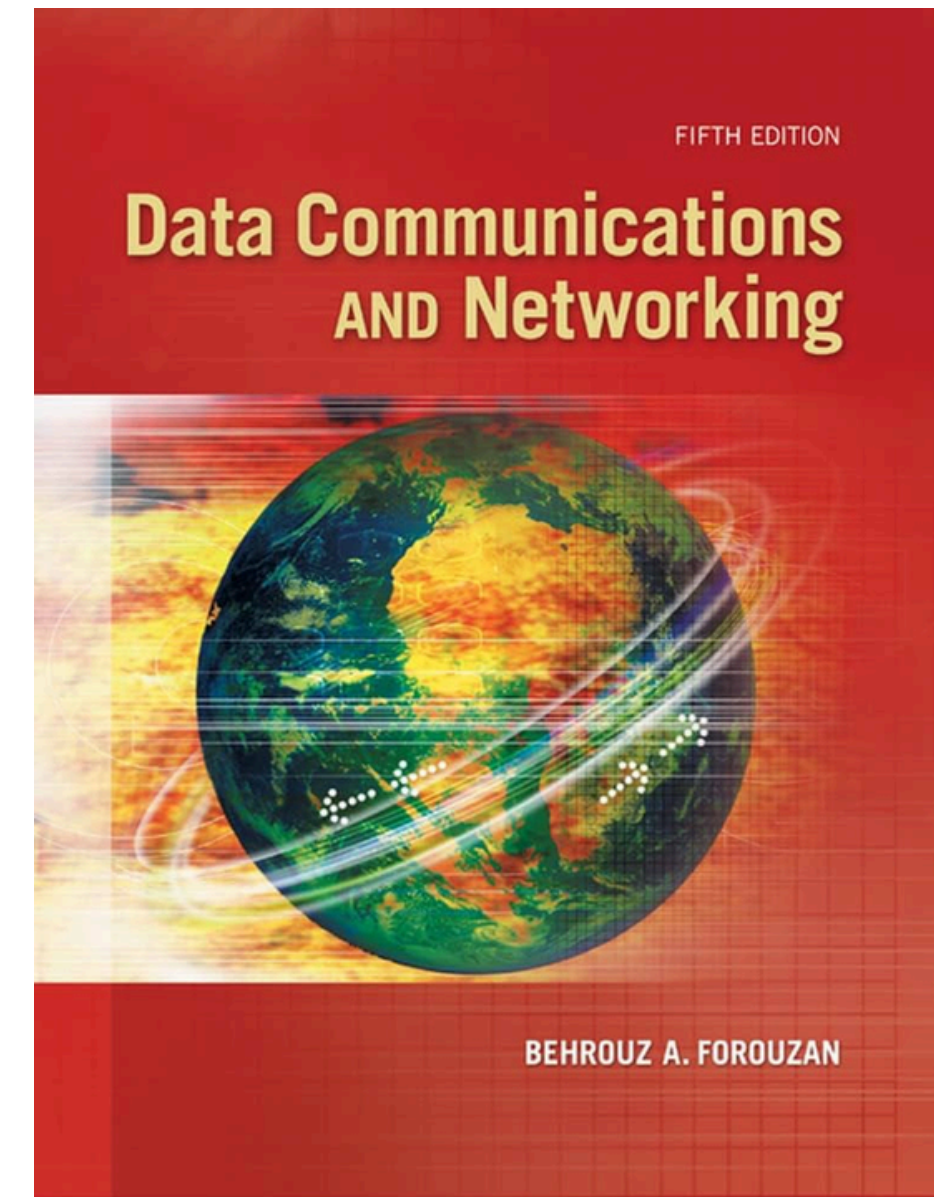
Contests	20%
Projects	20%
Mid Semester	20%
End Semester	40%

Reference Books

Reference Books



Computer Networking
James Kurose & Keith Ross



Data Communications and Networking
Behrouz A. Forouzan

Please fill in the feedback form.



**Thank
You!**